Foundation of Blockchain Technology (IS501)
Pre-Reading Material – Lecture 37

Merkle Tree

Estimated Reading Time: 15–20 minutes

Pre-Reading Material: Merkle Tree

1. Introduction

A Merkle Tree, also known as a Hash Tree, is a fundamental data structure used in blockchain technology to efficiently and securely verify the integrity of data.

It was introduced by Ralph Merkle in 1979. A Merkle Tree organizes data in a hierarchical structure in which individual data blocks are hashed, and their hashes are combined repeatedly to generate a single final hash called the Merkle Root.

Merkle Trees help blockchain networks verify transactions without requiring users to download and examine every transaction in a block.

2. Structure of a Merkle Tree

A Merkle Tree consists of the following components:

1. Leaf Nodes: The bottom-level nodes containing hashes of individual transactions.
2. Intermediate Nodes: These nodes are generated by combining and hashing the values of two child nodes.
3. Parent Nodes: Each parent node represents the combined hash of its two child nodes.
4. Merkle Root: The topmost hash representing the integrity of all transactions included in the tree.

3. Working Principle of Merkle Tree

The construction of a Merkle Tree follows these steps:

1. Individual transactions are collected in a blockchain block.
2. Each transaction is converted into a cryptographic hash using a hashing algorithm such as SHA-256.

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3. The transaction hashes become the leaf nodes of the tree.
4. Two adjacent hashes are concatenated and hashed together to generate a parent hash.
5. This process continues at each level until only one hash remains.
6. The final hash is called the Merkle Root.

The Merkle Root is stored in the block header and acts as a digital fingerprint of all transactions in that block.

4. Example of Merkle Tree

Consider a blockchain block containing four transactions:

- Transaction T1
- Transaction T2
- Transaction T3
- Transaction T4

Each transaction is independently hashed to generate four leaf hashes.

The hashes of T1 and T2 are combined to generate one parent hash. Similarly, the hashes of T3 and T4 generate another parent hash.

Finally, both parent hashes are combined and hashed to generate the Merkle Root.

5. Merkle Proof

A Merkle Proof is a cryptographic method used to verify whether a particular transaction belongs to a block without downloading all the transactions.

It requires only the transaction hash and a small number of intermediate hashes along the path to the Merkle Root.

Example: If a user wants to verify Transaction T1, the user does not need all four transaction records. The required sibling hashes can be used to reconstruct the Merkle Root and compare it with the trusted Merkle Root.

This makes transaction verification faster and more efficient.

6. Advantages of Merkle Tree

1. Data Integrity: Helps detect unauthorized changes in transactions.

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2. Efficient Verification: Allows transaction verification without downloading the complete block.
3. Reduced Storage Requirements: Supports lightweight blockchain clients.
4. Security: Uses cryptographic hash functions to protect transaction data.
5. Scalability: Makes verification efficient even when blocks contain large numbers of transactions.

7. Applications in Blockchain

Merkle Trees are widely used in:

- Bitcoin transaction verification.
- Blockchain block-header construction.
- Simplified Payment Verification (SPV).
- Ethereum data structures and transaction verification.
- Distributed storage systems.
- Secure data integrity verification.

8. Conclusion

Merkle Trees are an important cryptographic data structure in blockchain technology. They organize transaction hashes in a hierarchical structure and generate a single Merkle Root that represents the integrity of all transactions in a block. Their ability to support efficient verification through Merkle Proofs makes them an essential component of blockchain security and scalability.

Key Terms to Remember: Merkle Tree, Hash Function, Leaf Node, Parent Node, Merkle Root, Merkle Proof, SHA-256, Transaction Integrity, Simplified Payment Verification (SPV).

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